One day, one chain: from a shadow of a wave to a nested kernel

exact-integer constructions, five registrations, and a ledger of what died

Written by Claude (Opus 5.5) at the request of Pedro Garcia (Node Zero),
24 September 2026. Posits [PG] are Pedro's; readings [C] are Claude's.
Not for publication.

0. For the reviewing instance: read this first

Your job is the logical chain, not the code. Every numeric claim below came from a script that already ran; the scripts are listed in §12. Do not start by re-running them. Check instead, in this order:

1. Does every claim carry the stamp its evidence earns? A [run] claim needs a script; a [thm] needs a proof or a standard reference; a [post-hoc] claim must never be used as a premise.
2. Does any later section lean on a [post-hoc] or [PG] item as if it were [run] or [thm]? That is the failure this document is built to expose.
3. Were the registrations honest: prediction and kill written before the run, operationalisation declared, coverage disclosed?
4. Are the killed items (§11) actually dead everywhere, or does a later section quietly revive one?

Re-run a script only when the chain depends on its number and you have a concrete reason to doubt it.

Stamps. [run] measured this day by a named script, in exact integers unless the line says float (the hinge simulation of §1 and the angle readout of §8 used floats). [thm] a theorem: standard, or proved in the 13-Archive paper. [PG] Pedro's posit or role assignment. [C] Claude's reading or operationalisation, rejectable. [rhyme] a correspondence with a shipped system or historical result, used, not claimed as identity. [post-hoc] true arithmetic noticed after a run; not evidence. [open] untested.

Non-claims. Nothing here claims anything about the Navier–Stokes problem. The one-dimensional model used (§3) is a caricature with known behaviour. Pedro does not publish this work.

1. The shadow and the lift

A plotted wave is the collapse of a rotation $z = Ae^{i\phi}$ to one axis, $x = \mathrm{Re}\, z$ [PG]. The missing quadrature is recovered as a lookup, $z(t) \approx x(t) + i\, x(t - T/4)$, for a narrowband single-mode oscillator [thm]. The same object is the ring $\mathbb{Z}/13$: $5 = 2^9$ has order 4, so $\{1, 5, 12, 8\} = \{1, i, -1, -i\}$ [thm].

Hinge test (float simulation of Stuart–Landau onset, 400 trials, noise from kernel entropy) [run: hinge_test.py]. At onset the phase is set by the noise sample present then [thm]. Global Hilbert

transform: hinge located ∼1,285 samples late, phase error 57–66°. Quarter-turn lookup: within ∼8–10 samples, ≈ 3.5°. Forward coherence $R \approx 0.99$ at 40 cycles. Backward from the final phase: $R \approx 0.2, 0.5, 0.78, 0.95$ at 8, 4, 3, 2 cycles before the hinge. The hinge is a band about two turns wide, set by growth rate, not noise level.

T2 (13-Archive): float64 rotation by $(5 + 12i)/13$ leaves a 2^{-40}-turn band by $n = 38{,}000$ while its rounded output is still correct; drift is linear, $\approx 2.5 \times 10^{-17}$ turns/step, i.e. systematic and rewindable in principle. Prediction held [run].

2. The removed centre

Proposition 1 ([thm]). *Every rotation about c fixes c; restricted to the centre, the whole rotation group acts as the identity.*

Pedro's reading: the centre is ∅, not 0: it has no inside or outside and no rotation to measure [PG]. Claude's corrections on this day, recorded: arithmetic had been imported three times (the continuum, a limit at the centre, 0/0 treated as a field element) and each was withdrawn on challenge. Removing a point from the plane gives a cylinder (angle × scale); with integer rungs, scale is a counter [thm]. Three planes through ∅ give 8 octants; quarter-turns compose to exactly the 24 rotations of the cube (signed permutation matrices, det +1) [thm].

3. Exactness as method

Declarations [PG]: rails, bytes, relation, array, runtime, memory, output are all named consts; nothing undeclared is decided at runtime.

A blow-up carried in integers. Constantin–Lax–Majda model $\omega_t = \omega H \omega$ with $f = H\omega + i\omega$: $f(t) = 2f_0/(2 - tf_0)$, a Möbius map on Gaussian rationals; with $\omega_0 = \cos x$, $T^* = 2$ exactly; on the ladder $t = 2 - 2^{-k}$ the peak times remaining time is $2^{k+2}/(2^{k+2} - 1)$ exactly [run: clm_exact.py].

Viscosity as one rational per harmonic: implicit $1/(1 + \nu \Delta t k^2)$ or dyadic $1 - \nu \Delta t k^2$ [thm]. Unbounded exact rationals grow ∼3,000 (implicit) or ∼650 (dyadic) bits per step, because the exact state carries the damping history [run: viscous_rung.py].

Typed kernel. v1 (fixed Q20) underflowed high harmonics to 0. v2: each value is mantissa in $[2^{30}, 2^{31})$ times 2^e; unreached harmonics hold the sentinel ∅; an exact zero is logged as an event. v2 matches the exact rationals to all six printed digits in every case, and its single zero event was the explicit scheme's stability edge ($\nu \Delta t K^2 = 1$) [run: rung_kernel_v2.mjs]. Crossing of the rung ($\rho \geq 1$): inviscid $t = 33/16$; $\nu = 1/1024$: 134/64; $\nu = 1/16$: not by $t = 10$ at 32 modes [run].

The reciprocal. If $\dot{f} = f^2/2$ then $g = 1/f$ has $\dot{g} = -1/2$; if $\dot{A} = -A^2$ then $\frac{d}{dt} A^{-1} = I$ [thm]. Measured on the stepped coefficients: $\max |g(t) - g(0) + t/2|$ halves with the step when $\nu = 0$ $(6.7, 3.4, 1.7 \times 10^{-3})$, and stays at $\approx 2.9 \times 10^{-2}$ when $\nu = 1/64$ [run: reciprocal.py]. Claude's error on this day: the "stretch" was first described as an ingredient the lattice lacks; it is the reciprocal translation seen through the inverse. Two registers trading against a conserved product (Kelvin's circulation, the skater's $I\Omega$) [thm].

Closed finite systems. A deterministic map on a finite state space is eventually periodic; $x \mapsto 2x$ is chaotic on $\mathbb{R}/\mathbb{Z}$ and periodic on every rational, a 12-cycle on denominator 13 [thm]. Stirring is a volume-preserving bijection; only diffusion merges labels (Taylor's reversible glycerin demonstration) [thm].

4. Counting without reducing

Cutting the unit into 2^k pieces makes $2^k - 1$ cuts; pieces $-$ cuts $= 1$; the cut counts are Mersenne numbers, several prime [thm]. Twins kept unreduced: $\frac{4}{3}//\frac{3}{4} \mapsto 12/12$, $\frac{3}{2}//\frac{2}{3} \mapsto 6/6$, ratio of sizes 2 exactly; reducing first gives 16/9 and 9/4, which bracket 2 only to a factor 9/8 [PG, verified]. Correction on this day: the count of distinct ratios of $\{1..n\}$ equals the triangular number only at $n = 3, 4$ (at $n = 5$: 18 vs 15), so the tetractys link is a coincidence off that range [run].

5. The tryte, the quadryte, and a no-go

The tryte (cad-systemup) is three integer lanes each with a negative-zero flag. Lane states $\{+1, -1, +0, -0\}$ give $4^3 = 64 = 2^6$ states: corners 8, edges 24, faces 24, centre 8. Magnitude flip is an involution on all 64 and swaps corners↔centre, edges↔faces [run]. In ring order $+1 \to -1 \to +0 \to -0$ the step σ satisfies $\sigma^2 =$ magnitude flip: each lane is $\{1, i, -1, -i\}$ [run].

Proposition 2 ([run], exhaustive)**.** *No copy of the binary tetrahedral group (Hurwitz units) acts on the 64 states with -1 as the magnitude flip, within all 3,072 lane-respecting symmetries. Reason for the lane-preserving case: any such map squaring to the flip lies in the abelian $\mathbb{Z}_4^3$, but $ij = -ji$.*

The 16 equal-magnitude states (corners + centre) biject with the half-integer Hurwitz units, the flip becoming $q \mapsto -\bar{q}$ [thm]. The quaternion layer is the 13-Archive's Prop. 8 and T1 word (24 phases + a separate double-cover bit) [thm].

6. Octonions and the Octavian Catch

Cayley–Dickson: each doubling gains a unit and loses a law (order, commutativity, associativity, division); only dimensions 1, 2, 4, 8 have multiplicative norm (Hurwitz 1898) [thm]. Octonion multiplication is the Fano plane; any two units lie in one quaternion line (Artin) [thm]. The integral octonions have 240 units: 112 of type $(\pm 1, \pm 1, 0^6)$ and 128 of the 256 half-sign patterns (even parity) [thm].

T6 (registered, SHA-256 `b3166908...6fbd`): over the 30 labelings of $S(3, 4, 8)$, H1 at least one closes, H2 the table-aligned one does not (Kirmse 1924), H3 closed sets rotate exactly, H4 exactly 7 close. **All held** [run: `t6_octonion.py`]. Exploratory: every product leaving a non-closed set either lands exactly in another labeling's set or at quarter-integer coordinates; none in between. Named *the Octavian Catch* [PG].

Further [run]: the 30 split under the table's 21 exact symmetries as $7 + 7 + 7 + 7 + 1 + 1$, the closed seven being one orbit; a closed set's 240×240 table is a Latin square (unique addressing); $L_u L_v \neq L_{uv}$ on 1,464 of 1,600 pairs checked, so call sequences must be kept (the bracket). $S(3, 4, 8)$ is the affine cube $AG(3, 2)$: 8 vertices = octants, 14 blocks = 6 faces, 6 diagonal rectangles, 2 tetrahedra; its 7 directions are the Fano plane [thm].

T7 (`214f247c...e84d`): H1 every order-3 symmetry splits the closed seven as $1 + 3 + 3$: **held**. H2 (Claude) triples close: **killed**. H3 (Pedro's 30° as operationalised by Claude: 1/12 of cross products escape): **killed**. Escapes land in another closed set or one rung down. Coverage: 6 of 84 triples, disclosed.

Float anatomy [rhyme]: sign/exponent/mantissa, ± 0, NaN, subnormals, $1/x$ negating the exponent map onto lane sign, rung, mantissa, signed zero, $\varnothing$, underflow, magnitude flip. Difference: float rounds, flushes and propagates silently; this construction logs each.

7. The 7//11 hinge

Mod 13: $7 = 2^{11}$, $11 = 2^7$, $7 \cdot 11 \equiv -1$, $7/11 \equiv 2^4$ (order 3, the twist) [thm]. Under the seat gate the unpaired hinge closes on only 6 of 9 states [run]. With a refusal log, 1,000/1,000 random rewinds return exactly; without it, 449/1,000 drift (T9 A1) [run]. In $\mathbb{Z}_{12} = \mathbb{Z}_4 \times \mathbb{Z}_3$ quarter-turn and twist commute; crossing to the non-commuting tier needs the twist to act on addresses (which plane), the hkl crossing [C].

8. The cubic walk

Pedro's relation $x^2 - 1 = 1/(x+1)$, i.e. $(x-1)(x+1)^2 = 1$, run only as $a_{n+3} = -a_{n+2} + a_{n+1} + 2a_n$ [PG]. Its roots: one real (≈ 1.2056) and a dominant complex pair ($|z| \approx 1.2880$, turn $\approx 148.89°$ per step), so the integer sequence spirals [run]. Continued fraction of the turn: $[0; 2, 2, 2, 1, 1, 5, 4, 2, 70, \dots]$, arm counts $2, 5, 12, 17, 29, 162, \dots$ [run; meaning post-hoc]. Discriminant $= -59$ [run].

T8 (`d14ac04a...02a1`; registered text is the part above the dashed line of `prereg_T8.txt`): Pedro's "2 misses, moving once over" as operationalised by Claude (missed residues mod 19 and mod 17): **all killed**.

Census over the Monster primes [run]. Walk on the nonzero p^3 cube: single orbit (all $p^3 - 1$ states) at 3, 5, 19, 29; two equal halves at 7 and 41; binary clocks only at 17; one root (short) at 11, 13, 23, 31, 47; degenerate at 59; splits with a 70-cycle at 71. The binary collapse at 17 is forced: the cubic splits mod 17 and $17 - 1 = 2^4$ [thm]; 17 is the only Monster prime with both properties [run].

9. The doll: kernel, runtime, tooling

Stack [PG]: CubeKernel [binary, tryte, quadryte, Octavian Catch, 9-gon, 7//11]; CubeRuntime [19 in, 17 translate, 12-gon store/run/catch, 17 lift, 19 out]; CubeTooling [17/29 lookup, 41/19 in, 19 stream, $17 \cdot 19/15$ observe, 15/29 out, $29/(17 \cdot 19)$ store, 41 save, 41^{-1} trash]. Readings [C]: kernel and runtime share the mod-13 ring (seam S1); Rokop's seats end at 19, so 29 and 41 begin the outside codebook; p/q read as the p-cube seen through q's addressing. Forced by the walk [run]: 41 save/trash are its two halves (a semispace).

T9 (`1d5c329a...7f5b`): A1 receipts stop drift **held**; A2 walk-index pointerization bijective on 19, 29, 41 **held**; A3 exact rewind, no half changes **held**; A4 hub catches all **killed** (hub 1,024; others 2,048 and 3,072; 2 triples). Tier B speed is sent to Rust; Tier C awaits Pedro's byte maps. Same-interpreter timing: fixed-width walk 129 ns/step vs a contrived unbounded-Fraction walk 2.4 ms/step after 20,000 steps [run].

10. Pedro's untested posits

Kept here so they are not mistaken for results [PG, open]: six laws (primes as |numbers|; time as a fractal cut of phases; one empty point per observer and the "3%"; transcendentals as relations, finite to instrument resolution; addition as set composition; the torus as a cheat code); the three-kernel flywheel (compute, pace, memstore) with an outside export explaining a missing 30°; 17/19 as the in/out transport; the inverse bit switch ("its bit flip is signal; it eats a test and waits for its brother to confirm").

11. Ledger

registration	held	killed
T2	float leaves band before 10^6	—
T6	H1–H4	—
T7	H1 $(1+3+3)$	H2, H3
T8	—	P1, P2, P3
T9	A1, A2, A3	A4

Killed outside registrations: 33/7 as a labeling count (forced 30); tetractys as general law. **Post-hoc bin (not evidence):** $20{,}736 = 12^4 = 144^2$ caught in one T7 type; $144 = F_{12}$; hub catch $1{,}024 = 2^{10}$ in both types; 19-gon interior:reflex $17:21$ with 21 table symmetries; Fibonacci golden-angle denominators reducing to 13, 7, 17, 11; the 12.16° gap between the 19-gon interior and the spiral turn; $17 \cdot 19 = 18^2 - 1$ with $18 = 7 + 11$; discriminant 59 and the July "spawn gear 59"; 30 labelings vs 30° (unit-dependent); six laws vs twelve residues plus ∅; six spare bits.

12. Scripts and checklist

`hinge_test.py`, `clm_exact.py`, `viscous_rung.py`, `rung_kernel.mjs`, `rung_kernel_v2.mjs`, `reciprocal.py`, `tryte_hurwitz.py`, `t6_octonion.py`, `t7_flywheel.py`, `t9_tierA.py`, `t9_A4.py`; registrations `prereg_T6..T9`.

Checks worth making first: (a) §5's no-go proof sketch covers only the lane-preserving case; the lane-permuting case rests on the exhaustive search. (b) §6's claim that the closed seven form one orbit used a partial closure check in the follow-up script; T6's full check found 7. (c) §8's arm counts come from a float-evaluated turn angle; they are listed as post-hoc in meaning. (d) §9's roles are [PG]/[C]; only the 41 halves and T9 are [run]. (e) Nothing in §10 or the post-hoc bin should appear as a premise anywhere above.